Abnormal Cholesterol in U.S. Young Adults:

ABSTRACT

CVD remains a leading cause of morbidity in the U.S. This study analyzes abnormal cholesterol prevalence (Total, HDL, Non-HDL) among young adults aged 18-26. Using NHANES (2021-2023) weighted data, we explore disparities by sex, BMI, and healthcare engagement behavior.

BACKGROUND

Cardiovascular disease economic costs exceed **$250 billion** annually. While screening is often emphasized for children, monitoring 18-26 year olds is critical for primordial prevention.

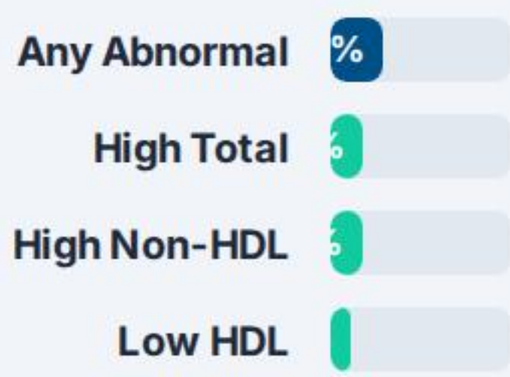
- High Total Cholesterol (≥200mg/dL)
- Low HDL-C (<40mg/dL)
- High Non-HDL-C (≥145mg/dL)

PRIMARY FINDINGS (NHANES 21-23)

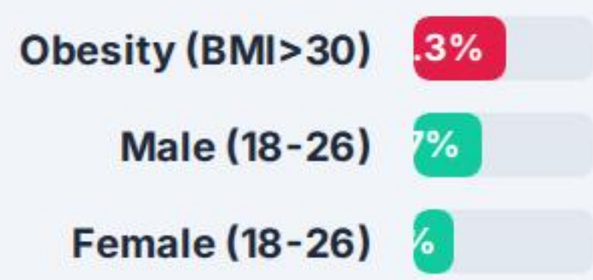
29.8% of Young Adults

Exhibit at least one abnormal cholesterol measure.

Weighted Prevalence Estimates (%)



Disparity by Subgroups



Young adults with obesity are over 2.5x more likely

METHODS

Design: Cross-sectional survey (August 2021-August 2023).

Analysis: Survey-weighted multivariable logistic regression. Accounts for multistage sampling design using **Taylor series linearization**.

Variables: Dependent (Abnormal Cholesterol); Independent (BMI, Sex, Smoking, Activity, Insurance).

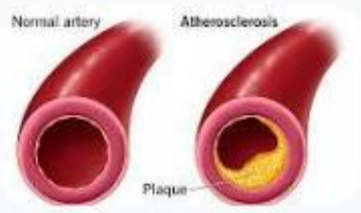
INTERPRETATION

Behavioral Impact: Smoking and sedentary behavior were strongly associated with low HDL. BMI serves as the most potent clinical predictor of dyslipidemia in the 18-26 cohort.

Access Barriers: RQ3 results suggest a significant gap between health literacy and actual phlebotomy results; those without a *usual source of care* had 1.4x higher odds of undiagnosed high non-HDL

Project: NHANES Analysis Poster | Investigator: Zhaoxing Li
Dimensions: 48" x 36" (Landscape Equiv.)

Image Sources



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